Answerable Biology — NGSS Alignment at a Glance

A one-sheet for administrators · high-school Biology, Grades 9–12 · Dr. Katherine von Duyke

Answerable Biology teaches each core idea through one **Variation Theory** arc: students commit to a first answer, work the idea through deliberate contrasts and a pattern-break, then revise. One AI feedback step turns that finished work into a one-page Conceptual Growth Report for each student and a class learning map for the teacher — individualized feedback without individualized grading. The program is three-dimensional by design, not by content coverage alone.

HOW THE PROGRAM MEETS THE STANDARDS

Three-dimensional NGSS. Each unit names its performance expectations and is built around the science and engineering practices and crosscutting concepts, not vocabulary recall.

Formative assessment, routine. The first-answer → revise → Growth Report loop makes thinking visible and reports growth, not just correctness (Black & Wiliam, 1998), and scales to 120 students.

Differentiation built in. Every concept question is written at three self-selected levels — Getting Started / Working On It / Mastery — so every student has an entry point (Vygotsky; Tomlinson).

Accessible by design. Questions use plain language so the difficulty lives in the biology, not the sentence — load-bearing for multilingual and Title I classrooms (Abedi & Lord, 2001).

Grounded, not improvised. The sequence follows Marton's Variation Theory; the feedback and proficiency reporting follow Marzano's *New Art and Science of Teaching* (2017).

UNIT-BY-UNIT ALIGNMENT

UNIT (CYCLE)	NGSS PERFORMANCE EXPECTATIONS	SCIENCE & ENGINEERING PRACTICES	CROSSCUTTING CONCEPTS
Ecosystems & Feeding Relationships (2)	HS-LS2-1, HS-LS2-2	Developing & using models; constructing explanations	Energy & matter; systems & system models; patterns
Energy Flow & Trophic Pyramids (3)	HS-LS2-3, HS-LS2-4	Mathematical & computational thinking; models; explanations	Energy & matter; cause & effect; scale, proportion & quantity
Cycles of Matter: The Carbon Cycle (4)	HS-LS2-3, HS-LS2-4, HS-LS2-5	Models; explanations; argument from evidence	Energy & matter; systems; stability & change
Photosynthesis (5)	HS-LS1-5	Models; explanations; argument from evidence	Energy & matter; structure & function; cause & effect
Cellular Respiration & Fermentation (6)	HS-LS1-7	Developing & using models; constructing explanations	Energy & matter; cause & effect; systems
Population Ecology & Succession (7)	HS-LS2-1, HS-LS2-2, HS-LS2-6 †	Analyzing & interpreting data; constructing explanations; models	Cause & effect; stability & change; systems
Cells & Organelles (8)	HS-LS1-2 †	Developing & using models; constructing explanations	Structure & function; systems & system models
Cell Membrane & Transport (9)	HS-LS1-2, HS-LS1-3	Developing & using models; constructing explanations	Structure & function; stability & change
Enzymes (10)	HS-LS1-6	Constructing explanations; models (lock-and-key); analyzing data	Structure & function; stability & change; cause & effect
The Cell Cycle → Cancer (11)	HS-LS1-4	Constructing explanations; analyzing & interpreting cases	Cause & effect; stability & change; systems
Reproduction & Meiosis (12)	HS-LS3-2 †	Developing & using models; constructing explanations	Cause & effect; structure & function; patterns
Mendelian Genetics & Punnett Squares (13)	HS-LS3-1, HS-LS3-3 †	Using mathematics (probability); explanations; analyzing data	Cause & effect; patterns; scale & proportion
DNA Structure & Replication (14)	HS-LS1-1, HS-LS3-1	Developing & using models; constructing explanations	Structure & function; patterns
Genes & Chromosomes (15)	HS-LS3-1 †	Developing & using models; constructing explanations	Structure & function; patterns
Mutations & Genetic Change (16)	HS-LS3-2, HS-LS3-1 †	Constructing explanations; models; argument from evidence	Cause & effect; structure & function; stability & change
Darwin & the Evidence for Evolution (17)	HS-LS4-1, HS-LS4-2 †	Analyzing & interpreting data; explanations; argument	Patterns; cause & effect; stability & change
Natural Selection & Adaptation (18)	HS-LS4-2, HS-LS4-3	Analyzing & interpreting data; explanations; argument	Cause & effect; patterns; stability & change
Speciation & Biodiversity (19)	HS-LS4-4, HS-LS4-5	Developing & using models (cladograms); explanations	Patterns; cause & effect; systems
Human Impact Capstone (20)	HS-LS2-7, HS-LS4-5, HS-LS4-6	Argument from evidence; explanations & designing solutions; data	Cause & effect; stability & change; systems; energy & matter

† Cycles 7, 8, 12, 13, 15, 16, and 17 performance expectations assigned by topic (not yet stated in the deck) — confirm before distribution. Performance expectations follow the Next Generation Science Standards (NGSS Lead States, 2013). References: Black & Wiliam (1998); Abedi & Lord (2001); Marzano, *The New Art and Science of Teaching* (2017); Marton, *Necessary Conditions of Learning* (2015).